

ALTERNATIVE 6.1

Lemma 0.1. *Assume (3.2)_n and (3.3)_{n-1}. Let (X, Δ) be a projective log smooth purely log terminal pair of dimension n , where $\Delta = S + A + B$, $\lfloor \Delta \rfloor = S$ is irreducible, A is a general ample $\mathbb{R}$ -divisor and B is an effective $\mathbb{Q}$ -divisor. Assume that $K_X + \Delta$ is pseudo-effective and $S \not\subset N(K_X + \Delta)$, then $\kappa(K_X + \Delta) \geq 0$ and if $\kappa(K_X + \Delta) < n$, then A is $\mathbb{Q}$ -Cartier.*

Proof. For all $t > 0$, $K_X + \Delta + tA$ is big and so this gives us a minimal model $\phi_t : X \dashrightarrow Y_t$. Note that $N_t := N(K_X + \Delta + tA)$ is given by the divisors contracted by ϕ_t . Since $N_t \subset N_{t'} \subset N_0$ for all $0 < t' < t$, one has that there exists a number $t_0 > 0$ such that $\text{Supp}(N_t) = \text{Supp}(N_0) =: N$ for all $0 < t \leq t_0$. We remark that S is not contracted by ϕ_t and we let $S_t = (\phi_t)_* S$. We have that for all $0 < t', t \leq t_0$, the rational maps $\phi_{t'} \circ (\phi_t)^{-1} : Y_t \dashrightarrow Y_{t'}$ are small.

Let $W = W_{t,t'}$ be a common log resolution of $X, Y_t, Y_{t'}$ admitting morphisms $p = p_{t,t'} : W_{t,t'} \rightarrow X$, $q_t = q_{t,t'} : W_{t,t'} \rightarrow Y_t$ and $q_{t'} = q_{t',t} : W_{t,t'} \rightarrow Y_{t'}$. We may write

$$q_{t,t'}^*(K_{Y_t} + S_t + (1+t')A_t + B_t) = q_{t',t}^*(K_{Y_{t'}} + S_{t'} + (1+t')A_{t'} + B_{t'}) + E_{t,t'}$$

where $E_{t,t'}$ is exceptional and $-E_{t,t'}$ is $q_{t,t'}$ -nef. By the Lemma 3.39 of [Kollár-Mori], we have that $E_{t,t'} \geq 0$ for all $t_0 \geq t, t' > 0$. Since $K_{Y_{t'}} + S_{t'} + (1+t')A_{t'} + B_{t'}$ is nef and big, it is semiample, so that $\kappa((K_{Y_{t'}} + S_{t'} + (1+t')A_{t'} + B_{t'})|S_{t'}) \geq 0$. It follows that

$$\kappa(S_t, (K_{Y_t} + S_t + (1+t')A_t + B_t)|S_t) \geq 0 \quad \forall t_0 \geq t, t' > 0$$

and hence that $(K_{Y_t} + S_t + A_t + B_t)|S_t$ is pseudo-effective. By (3.3)_{n-1}, then $\kappa(S_t, (K_{Y_t} + S_t + A_t + B_t)|S_t) \geq 0$.

Fix $m \gg 0$ such that

$$|\lfloor m(K_{Y_t} + \Delta_t) \rfloor - m(K_{Y_t} + \Delta_t)| < \epsilon$$

so that

$$m(K_{Y_t} + \Delta_t) - \lfloor m(K_{Y_t} + \Delta_t) \rfloor + A$$

is ample.

and $H^0(S_t, \mathcal{O}_{S_t}(m(K_{Y_t} + \Delta_t))) \neq 0$ (here $\Delta_t = S_t + A_t + B_t$). Let $t = 1/(m-1)$. We have a short exact sequence of reflexive sheaves

$$0 \rightarrow \mathcal{O}_{Y_t}(m(K_{Y_t} + \lfloor \Delta_t \rfloor) - S_t) \rightarrow \mathcal{O}_{Y_t}(m(K_{Y_t} + \lfloor \Delta_t \rfloor)) \rightarrow \mathcal{O}_{S_t}(m(K_{Y_t} + \lfloor \Delta_t \rfloor)) \rightarrow 0.$$

Since $\lfloor m(K_{Y_t} + \Delta_t) \rfloor - S_t$ is an integral Weil $\mathbb{Q}$ -Cartier divisor, (Y_t, B_t) is klt

$$\lfloor m(K_{Y_t} + \Delta_t) \rfloor - S_t - (K_{Y_t} + B_t) = (m-1)(K_{Y_t} + S_t + \frac{m}{m-1}A_t + B_t),$$

is nef and big, then by Kawamata-Viehweg vanishing (cf. (1-2-5) and (1-2-6) of [KMM87]), $H^1(Y_t, \mathcal{O}_{Y_t}(m(K_{Y_t} + \Delta_t) - S_t)) = 0$. The assertion now follows easily as the above sequence is exact on global sections and so $h^0(Y_t, \mathcal{O}_{Y_t}(m(K_{Y_t} + \Delta_t))) > 0$. Notice that as $(Y_t, \Delta_t + tA_t)$ is a minimal model for $(X, \Delta + tA)$, then we have $p^*(K_X + \Delta + tA) = q^*(K_{Y_t} + \Delta_t + tA_t) + E$ where E is effective and q -exceptional (here $p = p_{t,t'} : W_{t,t'} \rightarrow X$ and $q = q_{t,t'} : W_{t,t'} \rightarrow Y_t$). Since $-F = p^*A - q^*A_t$ is q -nef and q -exceptional, then by Lemma 3.39 of [Kollár-Mori], we have that $F \geq 0$. Therefore,

$$p^*(K_X + \Delta) = q^*(K_{Y_t} + \Delta_t) + E + t(q^*A_t - p^*A) = q^*(K_{Y_t} + \Delta_t) + E + tF$$

and so one easily sees that $h^0(X, \mathcal{O}_X(m(K_X + \Delta))) > 0$ as required. $\square$

Remark 0.2. *We have not used the full statement of (3.3) $_{n-1}$, but only the fact that if $K_S + A + B$ is pseudoeffective, then $\kappa(K_S + A + B) \geq 0$.*

Corollary 0.3. *Assume (3.2) $_n$ and (3.3) $_{n-1}$. Let (X, Δ) be a projective $\mathbb{Q}$ -factorial purely log terminal pair of dimension n , where $\Delta = A + B$, A is a general ample $\mathbb{Q}$ -divisor and B is an effective $\mathbb{Q}$ -divisor. Assume that $K_X + \Delta$ is pseudo-effective, then $\kappa(K_X + \Delta) \geq 0$.*

Proof.

Claim 0.4. *We may assume that (X, Δ) is log smooth.*

Proof. Standard. $\square$

By (2.3.2), if there is a positive integer k such that kA is Cartier and

$$h^0(X, \mathcal{O}_X(m(K_X + \Delta) + kA))$$

is bounded (for all $m > 0$ sufficiently divisible), then $K_X + \Delta$ is numerically equivalent to $N(K_X + \Delta)$. Moreover, it follows that $\kappa(K_X + \Delta) = 0$ (cf. [Nakayama]). Therefore, we may assume that there exist k as above such that

$$h^0(X, \mathcal{O}_X(mk(K_X + \Delta) + kA)) > \frac{(kn)^n}{n!}.$$

Therefore, for a general point $x \in X$, we may find an effective divisor

$$C \sim_{\mathbb{Q}} m(K_X + \Delta) + A$$

with $\text{mult}_x C > n$. For any $t \in [0, m]$, consider

$$(t+1)(K_X + \Delta) = K_X + \frac{m-t}{m}A + B + t(K_X + \Delta + \frac{1}{m}A)$$

$$\sim_{\mathbb{Q}} K_X + \frac{m-t}{m}A + B + tC = K_X + \Delta_t.$$

By construction $K_X + \Delta_t$ has the following properties:

- (1) $K_X + \Delta_m$ is not log canonical,
- (2) $K_X + \Delta_0$ is klt,
- (3) Δ_t is of the form ample plus effective for all $t < m$.

We may now consider $\mu : X' \rightarrow X$ a log resolution of $K_X + \Delta_t$, and we let $K_{X'} + \Delta'_t = \mu^*(K_X + \Delta_t) + E_t$ where E_t, Δ'_t are effective with no common components, $\mu_*\Delta'_t = \Delta_t$ and E_t is exceptional. Let $0 \leq F \ll \text{Supp}(F)$ be an effective exceptional divisor such that $A' \sim_{\mathbb{Q}} \epsilon A - F$ is ample. For $0 \leq t \leq (1-\epsilon)m$, let $\Delta''_t = \Delta'_t - \epsilon(\mu^{-1})_*A + A' + F$. After subtracting off common components of $K_{X'} + \Delta''_t$ and $N(K_{X'} + \Delta''_t)$, we may assume that for $t \in [0, (1-\epsilon)m]$

- (1) $K_{X'} + \Delta''_{(1-\epsilon)m}$ is not log canonical,
- (2) $K_{X'} + \Delta''_0$ is klt,
- (3) Δ''_t is of the form ample plus effective for all $t \leq (1-\epsilon)m$,
- (4) if S is a component of $\text{LCS}(K_{X'} + \Delta''_t)$, then $S \not\subset N(K_{X'} + \Delta''_t)$.

Let $s < (1-\epsilon)m$ be the defined by

$$s = \inf\{t \text{ s.t. } K_{X'} + \Delta''_t \text{ is not klt}\}.$$

Possibly perturbing Δ , we may assume that $\Delta''_s = S + A'' + B''$ where $\lfloor \Delta_s \rfloor = S$, A'' is ample and B'' is effective. By Lemma 0.3, $\kappa(K_{X'} + \Delta_s) \geq 0$, and the claim follows easily. $\square$